

MATGEO Presentation: 4.11.31

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Problem Statement

Find the area of the region bounded by the line $y = 3x + 2$, the X axis and the ordinates $x = -2$ and $x = 1$.

Strategy

The total area is the sum of the areas of two triangles.

We first find the intersection points (vertices) of the boundary lines:

$$y = 3x + 2$$

$$y = 0 \text{ (the x-axis)}$$

$$x = -2$$

$$x = 1$$

Then, we calculate the area of the two triangles formed by these vertices.

Finding Vertex B (Intersection with x-axis)

Intersection of $3x - y = -2$ and $y = 0$. The augmented matrix is:

$$\begin{pmatrix} 3 & -1 & | & -2 \\ 0 & 1 & | & 0 \end{pmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_2} \begin{pmatrix} 3 & 0 & | & -2 \\ 0 & 1 & | & 0 \end{pmatrix} \\ \xrightarrow{R_1 \rightarrow R_1/3} \begin{pmatrix} 1 & 0 & | & -2/3 \\ 0 & 1 & | & 0 \end{pmatrix}$$

This gives the vertex $\mathbf{B} = \begin{pmatrix} -2/3 \\ 0 \end{pmatrix}$.

Finding Vertex D (Intersection with $x = -2$)

Intersection of $3x - y = -2$ and $x = -2$. The augmented matrix is:

$$\begin{aligned} \left(\begin{array}{cc|c} 3 & -1 & -2 \\ 1 & 0 & -2 \end{array} \right) &\xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 0 & -2 \\ 3 & -1 & -2 \end{array} \right) \\ &\xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left(\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & -1 & 4 \end{array} \right) \\ &\xrightarrow{R_2 \rightarrow -R_2} \left(\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & -4 \end{array} \right) \end{aligned}$$

This gives the vertex $\mathbf{D} = \begin{pmatrix} -2 \\ -4 \end{pmatrix}$.

Finding Vertex E (Intersection with $x = 1$)

Intersection of $3x - y = -2$ and $x = 1$. The augmented matrix is:

$$\begin{aligned} \left(\begin{array}{cc|c} 3 & -1 & -2 \\ 1 & 0 & 1 \end{array} \right) & \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 3 & -1 & -2 \end{array} \right) \\ & \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & -1 & -5 \end{array} \right) \\ & \xrightarrow{R_2 \rightarrow -R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 5 \end{array} \right) \end{aligned}$$

This gives the vertex $\mathbf{E} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$.

Summary of Vertices

The remaining vertices are the intersections with the x-axis:

$$\mathbf{A} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$$

$$\mathbf{C} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

These points form two triangles: $\triangle ABD$ and $\triangle EBC$.

Area Calculation

The total area is the sum of the areas of $\triangle ABD$ and $\triangle EBC$. Area of

$\triangle ABD$:

$$\text{ar}(\triangle ABD) = \frac{1}{2} \|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{D})\| \quad (3.1)$$

$$= \frac{1}{2} \left\| \begin{pmatrix} -4/3 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 4 \end{pmatrix} \right\| = \frac{8}{3} \quad (3.2)$$

Area of $\triangle EBC$:

$$\text{ar}(\triangle EBC) = \frac{1}{2} \|(\mathbf{E} - \mathbf{C}) \times (\mathbf{B} - \mathbf{C})\| \quad (3.3)$$

$$= \frac{1}{2} \left\| \begin{pmatrix} 0 \\ 5 \end{pmatrix} \times \begin{pmatrix} -5/3 \\ 0 \end{pmatrix} \right\| = \frac{25}{6} \quad (3.4)$$

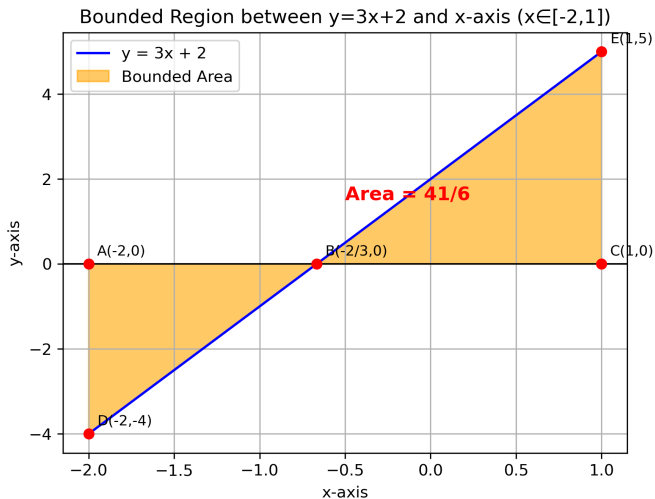
Total Area

The total area of the region is the sum of the two triangle areas.

$$\text{Area} = \text{ar}(\triangle ABD) + \text{ar}(\triangle EBC) \quad (3.5)$$

$$\text{Area} = \frac{8}{3} + \frac{25}{6} = \frac{16 + 25}{6} = \frac{41}{6} \quad (3.6)$$

Plot



Region bounded by the line and ordinates